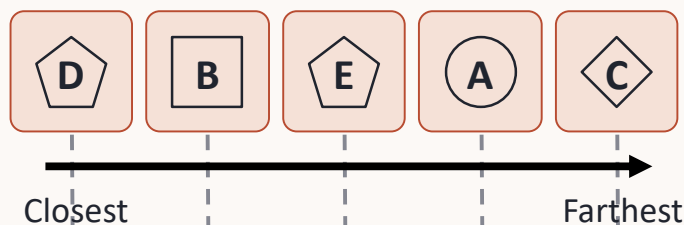
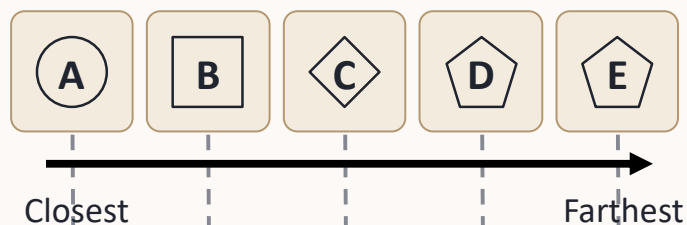


1 Learn action order before rewriting the representation.

Native predictive order
(by original terminal future-to-goal distance)

Decision-aligned relational order
(from complete-set relational evidence)



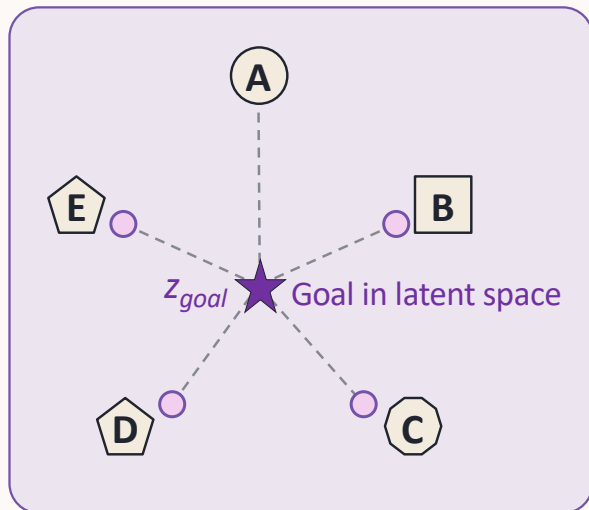
Relational target (training-only supervision)

3 Exact Ordinal Realization

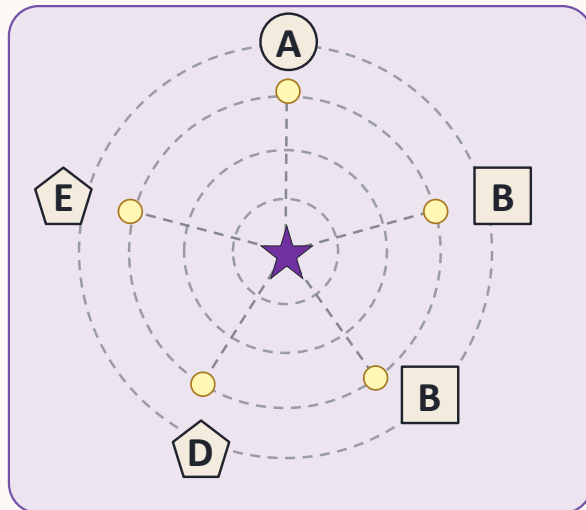
Encode the aligned order directly in future-to-goal distance.

Transported terminal futures (before)

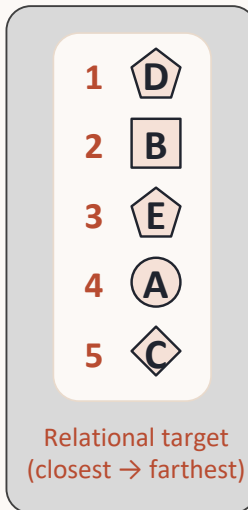
Realized terminal futures (after)



$target\ radius(a) \propto relational\ rank(a)$

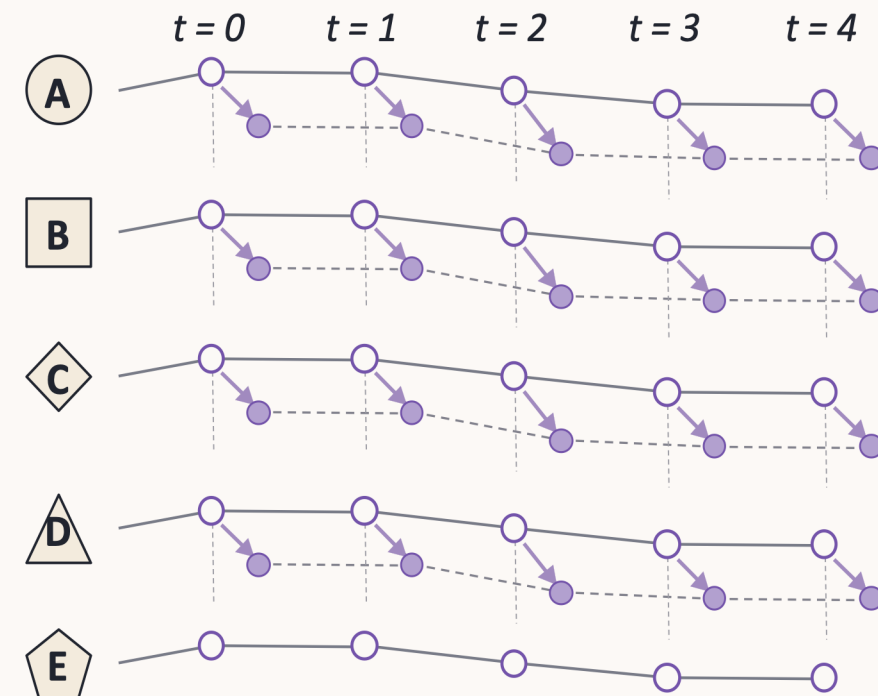
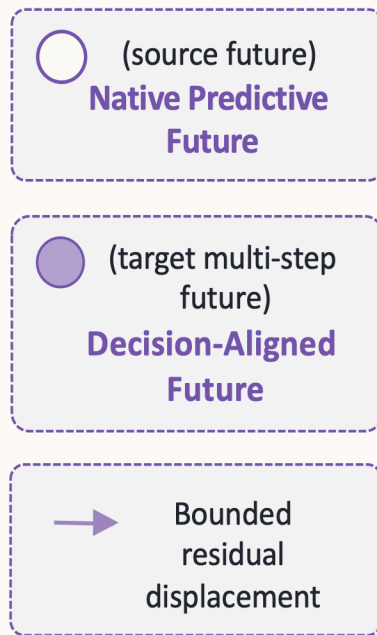


$rank\ d(z_{realized}(a), z_{goal}) = aligned\ relational\ rank$



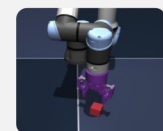
2 Bounded Temporal Transport

Learn a new multi-step future while preserving action correspondence.



4 Single-Checkpoint DA-JEPA

Observation (context)



...



Predict future JEPA representations (realized)

Native JEPA Criterion

$$a^* = \arg \min_a d(z_{future}(a), z_{goal})$$



Decision alignment lives in the future representation. One checkpoint. Native JEPA selection.

Candidate actions



Selected action